Ming Shin Cindy Sung

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) Sep 2023 – Jun 2025
MSc in Data-Centric Artificial Intelligence Technology (GPA: 4.06/4.30) Guangzhou, China

Modules: Foundation of Data Science (A+), Practical Lab Course (A+), Deep Learning (A)

Imperial College London Oct 2020 – Jun 2023
BSc in Medical Biosciences (Upper 2:1) London, UK

PUBLICATIONS & MANUSCRIPTS

M. S. C. Sung, et al. "LIBERO-Industry: Benchmarking Robotic Manipulation under Industrial Process Constraints." *arXiv:2610.16782*, 2026. *Submitted to ICLR 2027.* **Best Paper Candidate Award** (International Conference on Engineering Intelligence 2026, Workshop). [Project Page]

C. Tan, **M. S. C. Sung**, et al. "Human-Like Redundancy Resolution for Real-Time Bimanual Humanoid Teleoperation." *arXiv:2610.16855*, 2026. *Submitted to ICRA 2027.* [Project Page]

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant (Embodied AI) – Full-Time Apr 2026 – Present
Tongji University (Supervisor: Prof. Shuxian Liang, Prof. Xian-Sheng Hua) Shanghai, China

- **Benchmark & Evaluation:** Designed and curated a 216-task robot-learning benchmark under industrial process constraints; evaluated five policies across 1,800 held-out rollouts. Conducted cumulative stress testing, identifying a sharp decline in $\pi_{0.5}$ task success from 43.8% to 2.1% at the transition to multi-object manipulation.
- **Dataset & Infrastructure:** Built deterministic task-generation and MuJoCo/robosuite mechanics for moving material flows, with scripted demonstration collection, replay validation, and evaluation tooling; produced 1,412 simulation and 95 physical demonstrations for fine-tuning vision-language-action policies.
- **Real-World Validation:** Constructed an AUBOi5H conveyor-transfer workcell and conducted matched $\pi_{0.5}$ sim–real co-training experiments; the LIBERO-Industry condition showed a 12.5–34.7 percentage-point advantage in target-transfer completion over original-LIBERO across five physical variations.

AI Research Intern – Full-Time Feb 2025 – Mar 2026
Terminus Group (Supervisor: Dr. Shuxian Liang) Shanghai, China

- **VLA Efficiency Profiling:** Profiled $\pi_{0.5}$ across 10 LIBERO-Goal tasks, measuring latency, GPU memory/utilization, and component-wise FLOPs; identified action generation as the dominant inference-time cost.
- **Attention & Representation Analysis:** Developed layerwise representation-similarity and attention-sparsity diagnostics for Gemma-2B, using the results to prioritize decoding-side computation over KV-cache compression for efficiency optimization.

NLP & Data Analytics Intern Jul 2024 – Jan 2025
Huawei Technologies Co., Ltd. Shenzhen, China

- **Text-to-SQL Evaluation & Tooling:** Evaluated LLM-generated SQL accuracy and error patterns using PostgreSQL and MySQL; developed an automated SQL annotation tool, improving annotation efficiency by 20%.

SELECTED RESEARCH PROJECTS

Parameter-Efficient LLM Fine-Tuning – MSc Group Project Feb 2024 – Jun 2024
The Hong Kong University of Science and Technology (Guangzhou)

- Fine-tuned Llama-2-7B for question answering using LoRA, QLoRA, and prompt tuning; compared accuracy, inference speed, and storage requirements across adaptation strategies.

TECHNICAL SKILLS

Robotics & Simulation Frameworks: MuJoCo, robosuite, LIBERO, AUBOi5H + CTAG2F90D Deployment
Robot Learning & Efficiency: PyTorch, Imitation Learning, Inference Profiling
Programming & Tools: Python, SQL, R, MATLAB, Git, LaTeX

RESEARCH INTERESTS

Embodied Intelligence, Robot Learning Benchmarks, Model & Resource Efficiency, Smart Manufacturing Automation, Sim-to-Real Deployment